

Introduction to IoT and Embedded Systems

Module 2 Assignment

Here are two widely used embedded systems available on the market today, along with an analysis of their interfaces:

1. Arduino Uno

Overview:

Arduino Uno is a microcontroller-based development board based on the ATmega328P chip. It is commonly used in DIY electronics projects and educational applications.

Interfaces:

- **Digital I/O Pins:**
14 digital pins (of which 6 can be used as PWM outputs) allow connection with sensors, LEDs, relays, and other digital components.
- **Analog Inputs:**
6 analog input pins for connecting analog sensors like temperature or light sensors.
- **USB Interface:**
Used for programming the board and serial communication with a PC.
- **Power Interface:**
Can be powered via USB or external adapter (7–12V) through the DC power jack.
- **Serial Interface (UART):**
Allows for asynchronous serial communication with other devices.
- **SPI/I2C Interfaces:**
For communication with other microcontrollers, displays, or sensors.

Summary:

Arduino Uno provides a wide range of simple and accessible interfaces suitable for rapid prototyping and learning embedded development.

2. Raspberry Pi 4

Overview:

Raspberry Pi 4 is a single-board computer with an ARM-based processor. It runs a full Linux operating system and is used for a variety of IoT, robotics, and computing tasks.

Interfaces:

- **GPIO (General Purpose Input/Output):**
40-pin header with digital I/O pins that can be programmed for various purposes, including SPI, I2C, and UART communication.
- **USB Ports:**
Includes 2× USB 3.0 and 2× USB 2.0 ports for peripherals like keyboards, mice, flash drives, and cameras.
- **HDMI Ports:**
Dual micro-HDMI ports for display output (up to 4K resolution).
- **Ethernet Port:**
Gigabit Ethernet for wired internet connection.
- **Wi-Fi and Bluetooth:**
Built-in for wireless connectivity.
- **CSI and DSI Connectors:**
Camera Serial Interface (CSI) for camera modules and Display Serial Interface (DSI) for LCD displays.
- **MicroSD Slot:**
Used for operating system and file storage.

Summary:

Raspberry Pi 4 is a powerful embedded system that supports a wide range of high-speed, multimedia, and general-purpose interfaces. It bridges the gap between microcontroller boards and full-fledged desktop computers.

Here's a detailed breakdown of the **inputs and outputs** for each embedded system: **Arduino Uno** and **Raspberry Pi 4**.

1. Arduino Uno

Inputs:

1. **Digital Input Pins (0–13):**

- Can receive HIGH or LOW (1 or 0) digital signals.
- Used for push buttons, motion sensors, digital IR receivers, etc.

2. **Analog Input Pins (A0–A5):**

- Read analog voltages from sensors (0–5V), converted via 10-bit ADC.
- Used for temperature sensors, potentiometers, light sensors, etc.

3. **Power Input:**

- Via USB (5V) or external DC power jack (7–12V).
- Provides operating voltage to the board.

4. **Serial Input (UART):**

- Used to receive data from a computer or serial device.

5. **SPI and I2C Inputs:**

- Used for data input from compatible sensors or modules like an IMU or external ADC.

Outputs:

1. **Digital Output Pins (0–13):**

- Can output HIGH or LOW to control LEDs, relays, digital actuators, etc.

2. **PWM Outputs (Pins 3, 5, 6, 9, 10, 11):**

- Pulse-width modulated signals to control servo motors, dim LEDs, etc.

3. **Analog Output (via PWM):**

- Simulated analog signal using PWM (Arduino does not have true analog output).

4. **Serial Output (UART):**

- Sends data to the computer or another serial device.

5. SPI and I2C Outputs:

- Used to send data to displays, memory chips, or other microcontrollers.

2. Raspberry Pi 4

Inputs:

1. GPIO Input Pins:

- General-purpose digital input pins.
- Can read HIGH/LOW signals from buttons, switches, motion detectors, etc.

2. USB Ports:

- Accept input from keyboards, mice, webcams, USB sensors.

3. Ethernet Port:

- Receives network data for internet connectivity or LAN communication.

4. Wi-Fi & Bluetooth:

- Wireless data input from internet, sensors, or phones.

5. Camera Interface (CSI):

- Receives video input from Raspberry Pi camera modules.

6. MicroSD Card:

- Input source for operating system and storage.

7. Serial Interfaces (I2C, SPI, UART):

- Receive data from external microcontrollers or sensors.

Outputs:

1. **GPIO Output Pins:**

- Control LEDs, motors, relays, or communicate with other digital devices.

2. **USB Ports:**

- Send data to printers, flash drives, or USB-controlled hardware.

3. **HDMI Ports:**

- Video output to monitors or TVs.

4. **Audio Jack / HDMI Audio:**

- Output analog or digital sound to speakers or headphones.

5. **Ethernet / Wi-Fi:**

- Send data over networks or the internet.

6. **Serial Interfaces (I2C, SPI, UART):**

- Output data to displays, sensors, or other devices.

Here is a classification of the **inputs and outputs** for the **Arduino Uno** and **Raspberry Pi 4** based on their **mode of interaction**:

Arduino Uno

Inputs:

- **Tactile:**
 - Push buttons
 - Switches

- **Electronic:**

- Analog sensors (temperature, light, potentiometers)
- Digital sensors (IR sensors, motion sensors)
- Serial, I2C, SPI input data

Outputs:

- **Electronic:**

- LEDs
- Relays
- Servo motors
- Serial, I2C, SPI output data

Raspberry Pi 4

Inputs:

- **Visual:**

- Camera input via CSI
- USB webcam input

- **Tactile:**

- Keyboard
- Mouse
- Buttons or touch sensors via GPIO

- **Audio:**

- USB microphone (if connected)

- **Electronic:**

- GPIO digital signals
- I2C, SPI, UART data from sensors
- Ethernet/Wi-Fi data
- USB-connected sensors

Outputs:

- **Visual:**

- HDMI display
- GUI interface

- **Audio:**

- 3.5mm audio jack output
- HDMI audio output

- **Electronic:**

- GPIO output to devices (LEDs, motors)
- I2C, PI, UART communication
- USB data output
- Network transmission via Ethernet/Wi-Fi